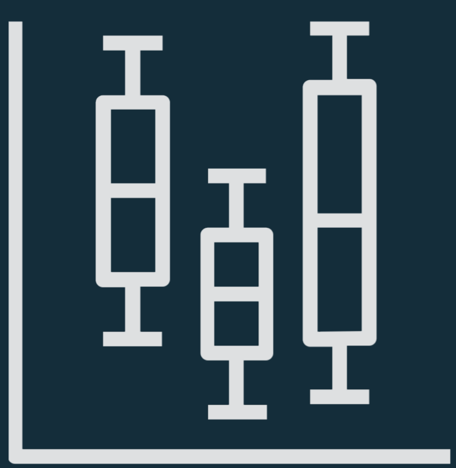
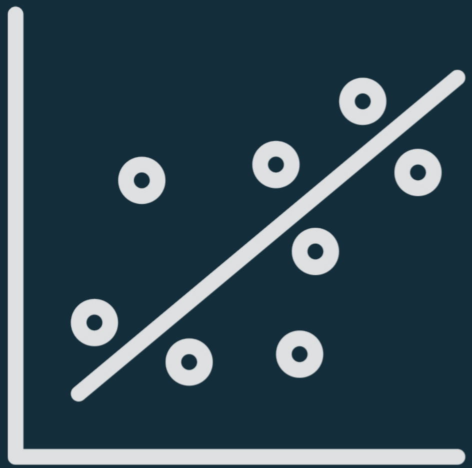
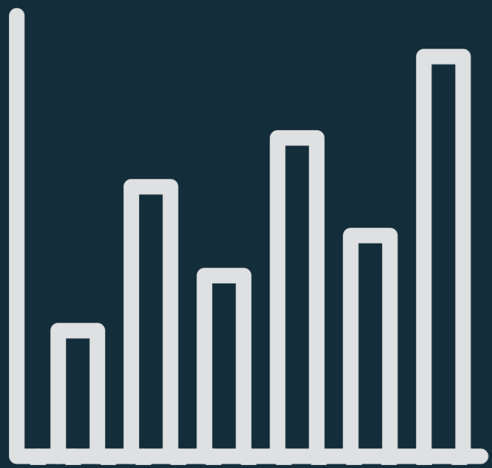
Data analysis







Slide prepared and presented by Ethel Pruss (PhD candidate at VU Amsterdam)

Learning Goals

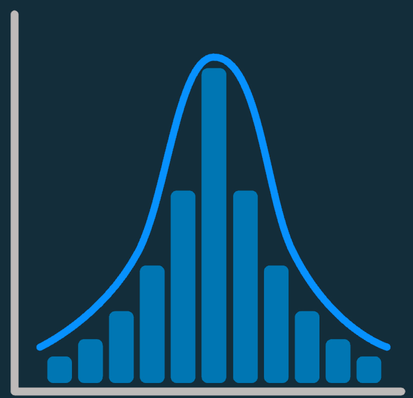
- A basic overview of:
  - Types of data
  - Measures of central tendency
  - Hypothesis testing
  - Visualization

- **Main goal:** knowing how we go from a hypothesis to results

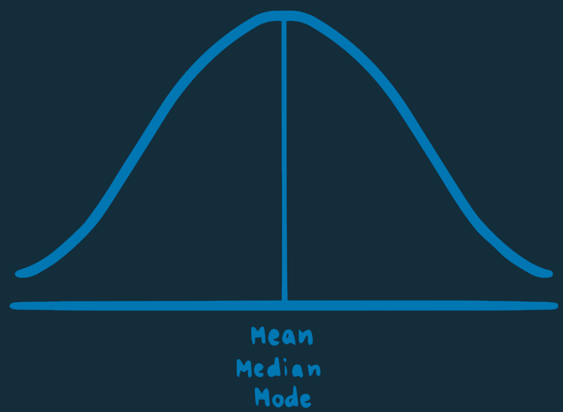
- *NOT* the mathematical implementation → we use tools

1.Descriptive Statistics

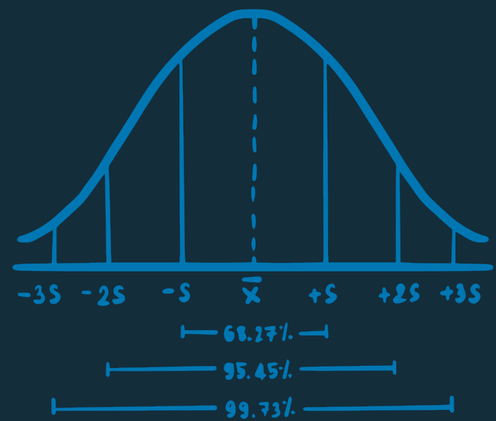
Describe the sample.



Distribution



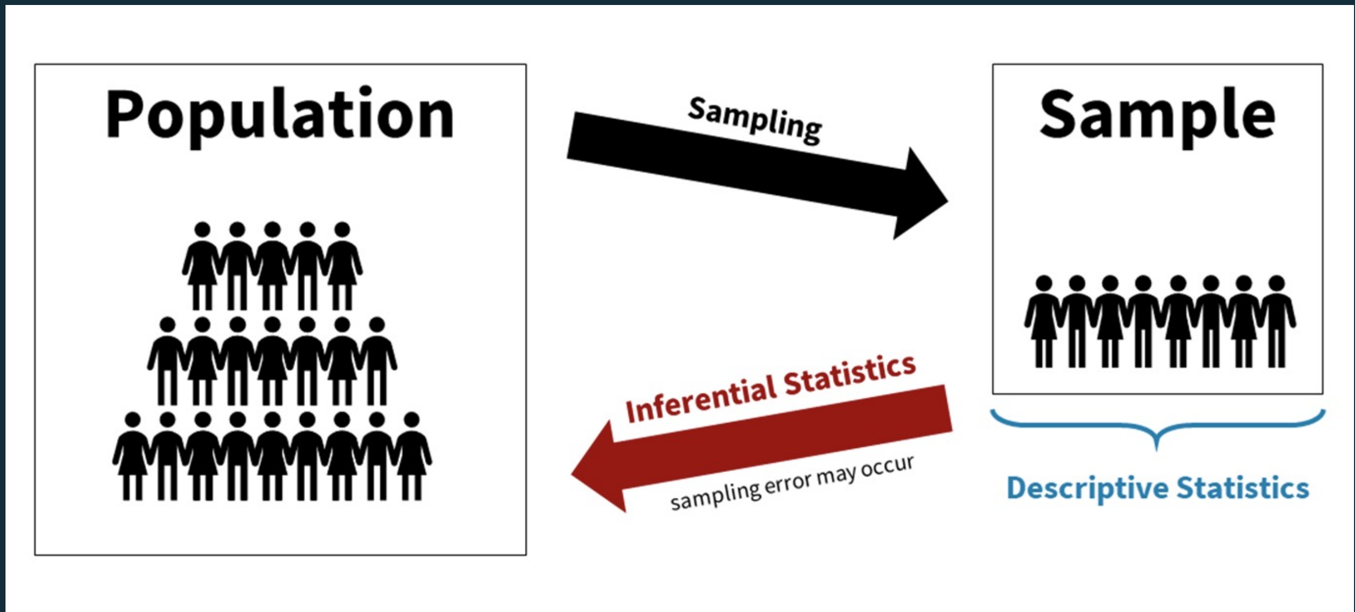
Central tendency



Variability

2. Inferential Statistics

Infer something about the population based on the sample.



Types of Data

Ratio

+ - * :

Equal intervals &
true zero point

Interval

+ -

Equal intervals &
arbitrary zero point

Types of Data

mean & sd

Ratio

\+ - * :

Equal intervals &
true zero point

Interval

\+ -

Equal intervals &
arbitrary zero point

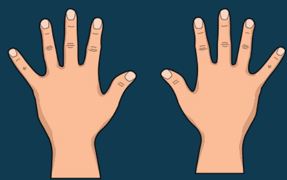
Types of Data

Ordinal

> < =

Ranking/order

Nominal



Frequency (counting), comparison (same or not)

Types of Data

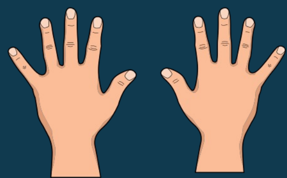
~~mean & sd~~

Ordinal

> < =

Ranking/order

Nominal



Frequency (counting),
comparison (same or not)

Likert Scales

Ordinal or interval?

TYPES	RESPONSE OPTIONS				
Agreement	**Stronly Agree**	**Agree**	**Neutral**	**Disagree**	**Strongly Disagree**
Likehood	Very Likely	Likely	Neutral	Unlikely	Very Unlikely
Satisfaction	Extremely Satisfied	Satisfied	Neutral	Dissatisfied	Extremely Dissatisfied
Quality	Excellent	Above Average	Average	Below Average	Poor
Frequency	Very Often	Often	Sometimes	Rarely	Never
Importance	Essential	Very Important	Of Average Importance	Of Little Importance	Not at all Important
Numeric	5	4	3	2	1

Ratio

+ - * :

Equal intervals & true zero point

Interval

+ -

Equal intervals & arbitrary zero point

Ordinal

> < =

Ranking/order

Nominal

Frequency (counting), comparison (same or not)

Examples

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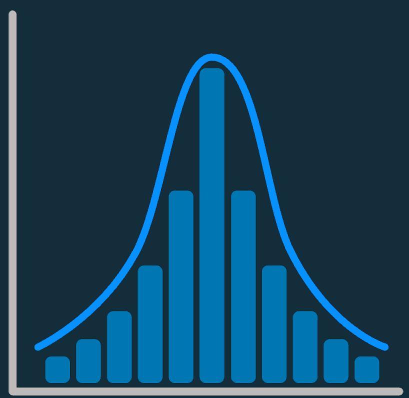
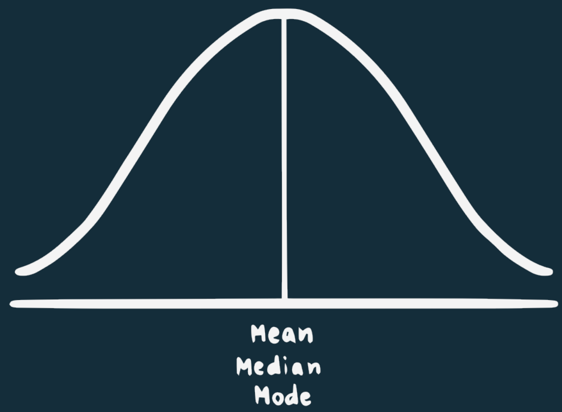
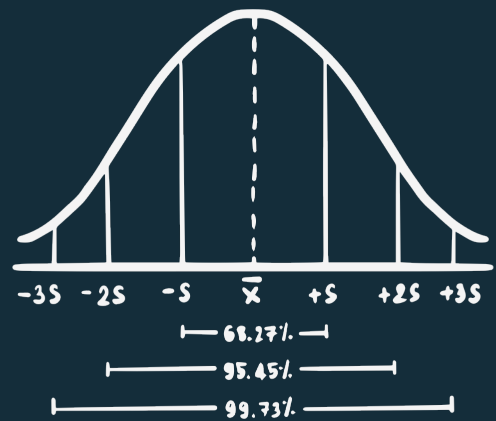
Descriptive Statistics



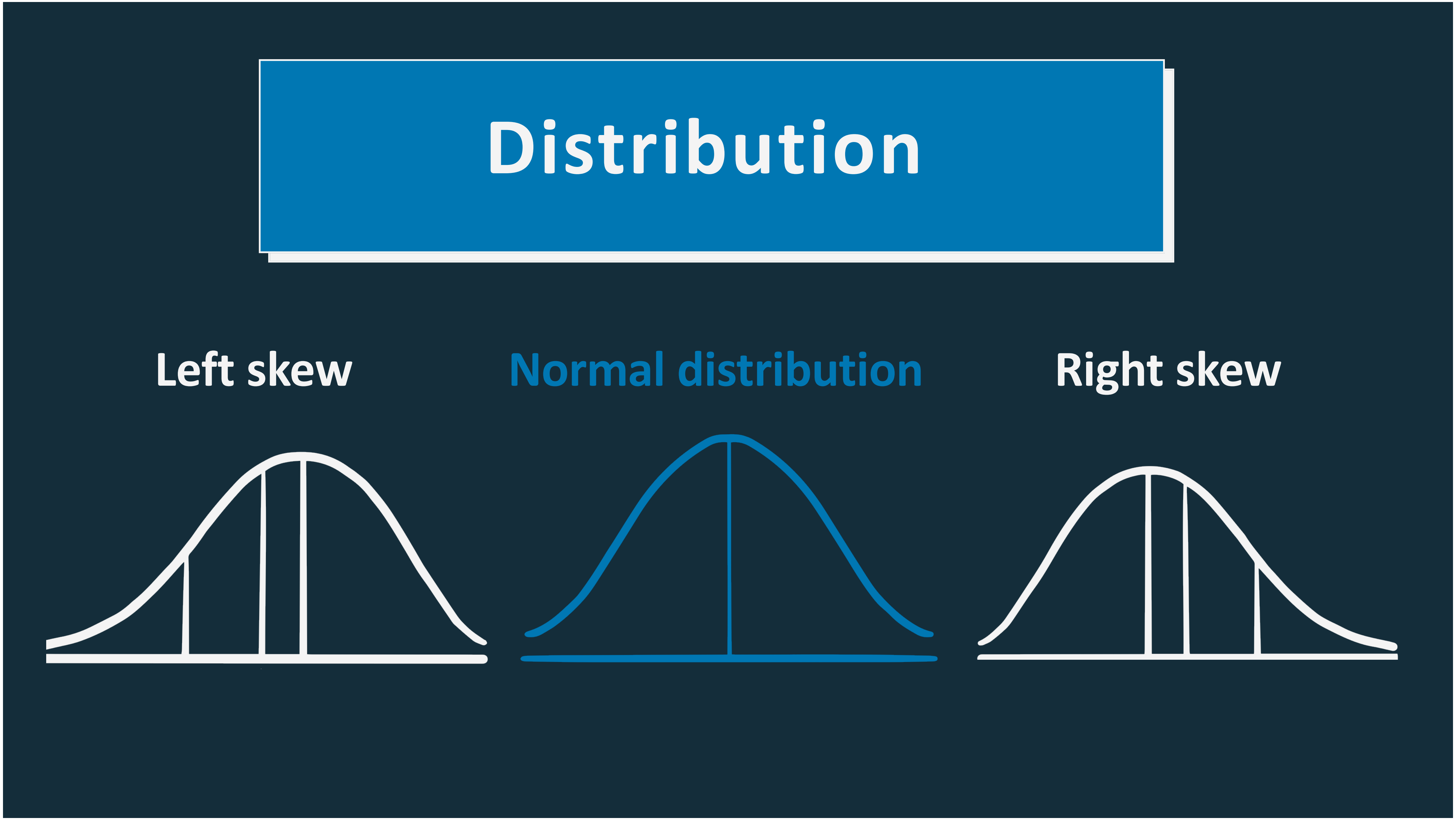
Distribution



Central tendency

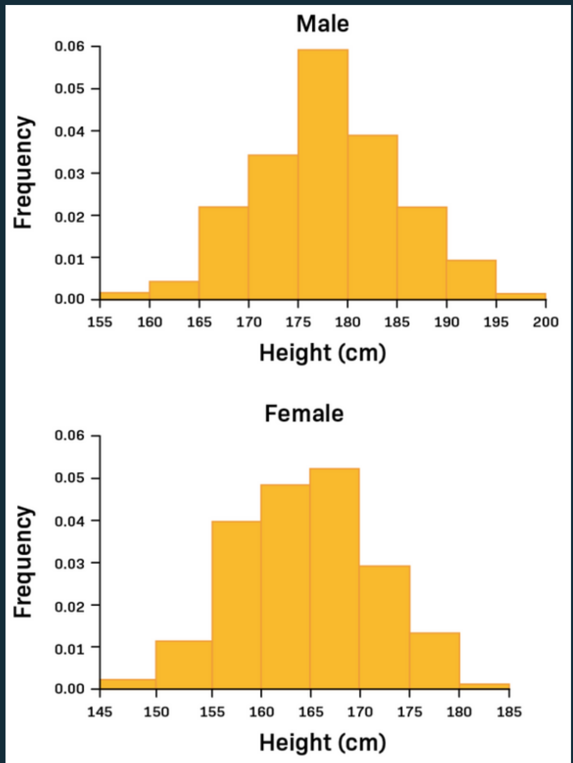


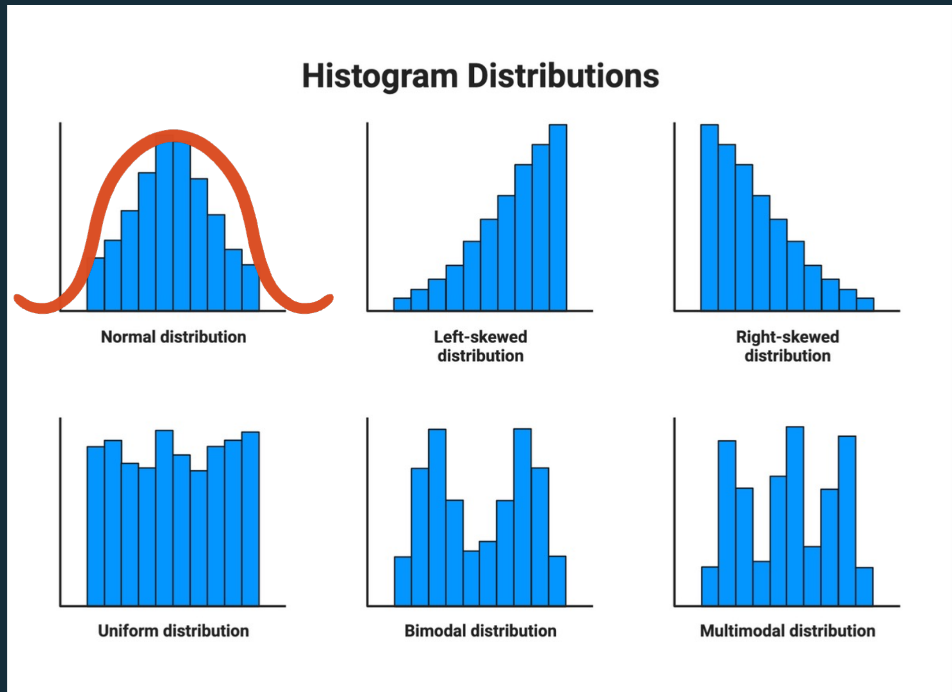
Variability


Distribution
Left skew
Normal distribution
Right skew

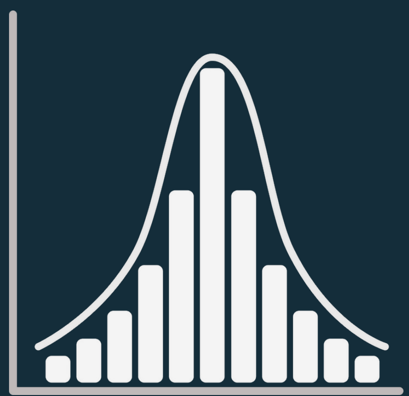
Distribution

Histograms



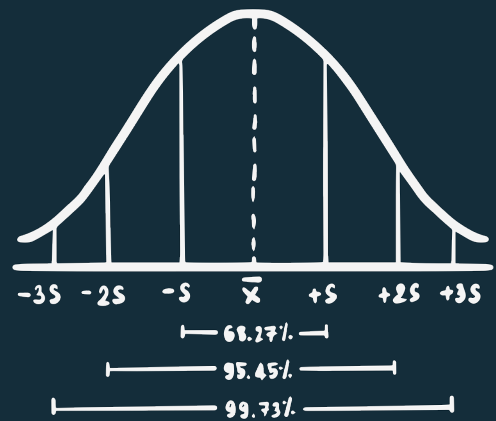


Descriptive Statistics



Distribution

Central tendency

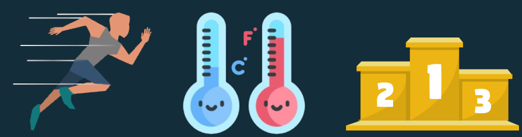


Variability

Measures of Central Tendency

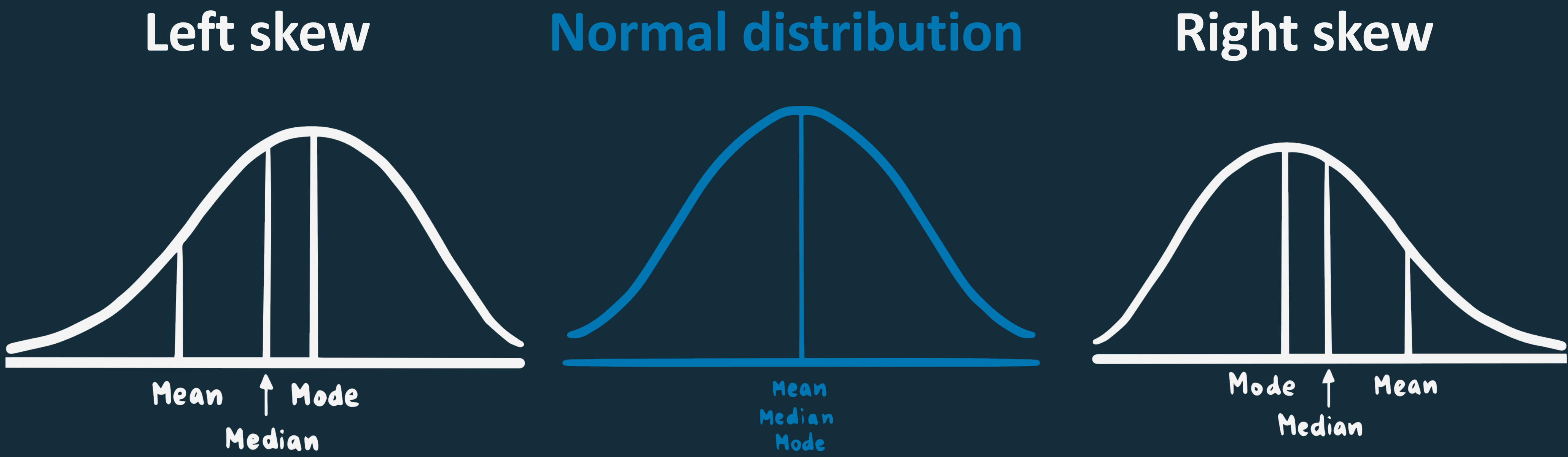
Mean: Sum of all values / total number of values

Median: Middle value in ordered data



Mode: Most common value

Measures of Central Tendency



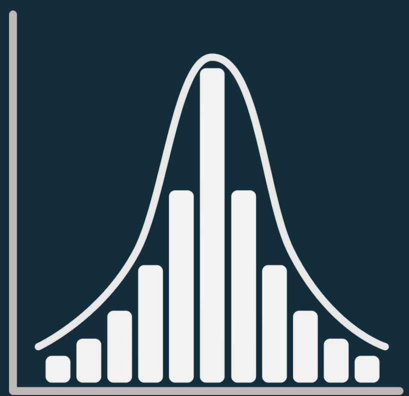
Pros & Cons

Mean: most accurate summary of data *but* heavily affected by outliers

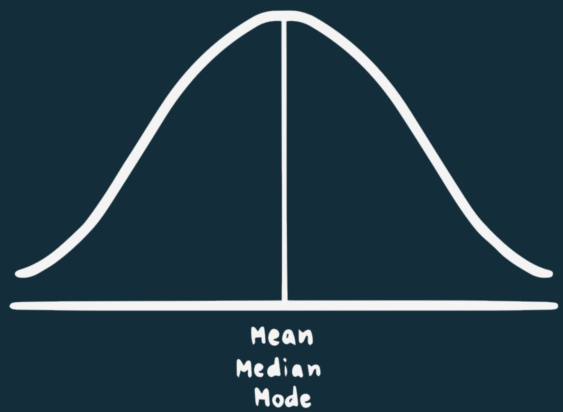
Median: relatively unaffected by outliers *but* doesn't represent all data

Mode: can be used with nominal data *but* doesn't represent all data & you can have multiple!

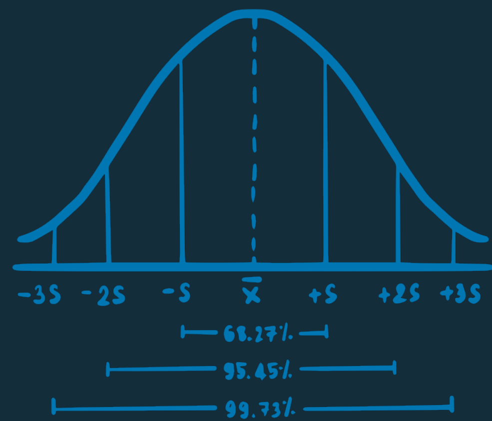
Descriptive Statistics



Distribution



Central tendency

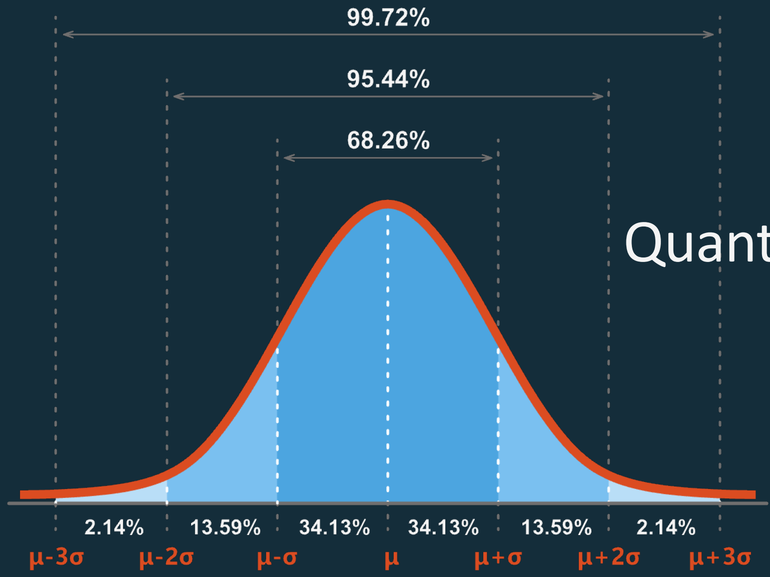


Variability

Measures of Variability

Confidence interval

- Range
- Standard deviation
- Variance
- Interquartile range



Quantiles

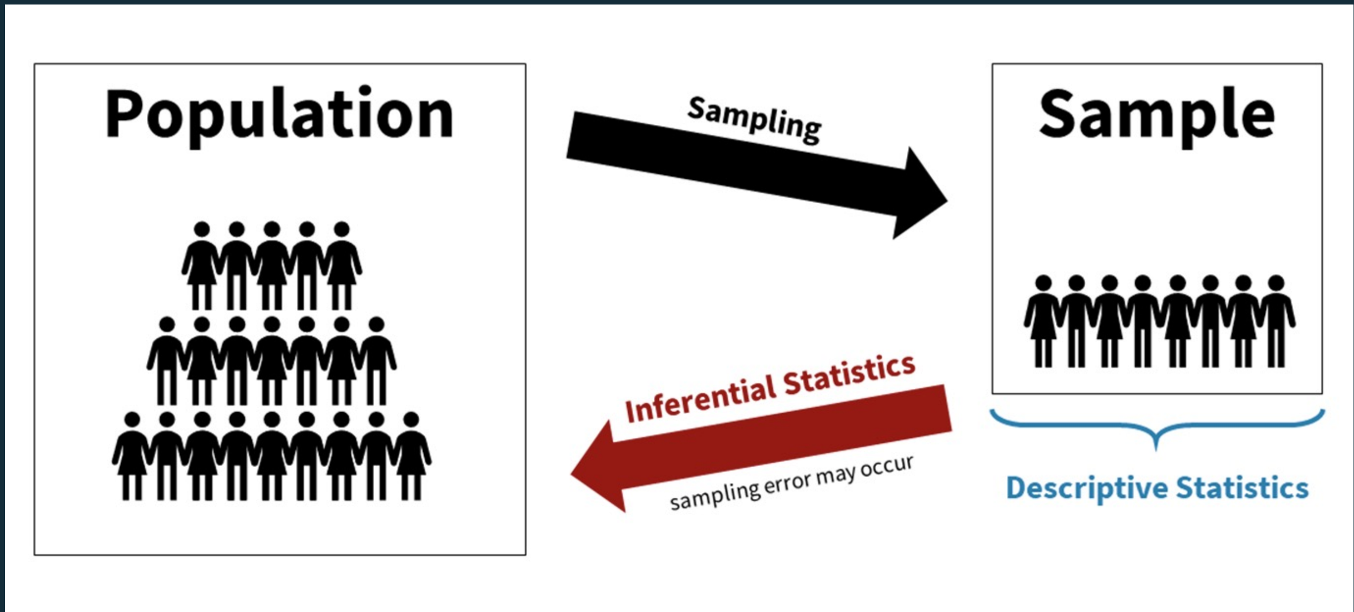
Standard deviations

15 Minute Break

Continuing at ...

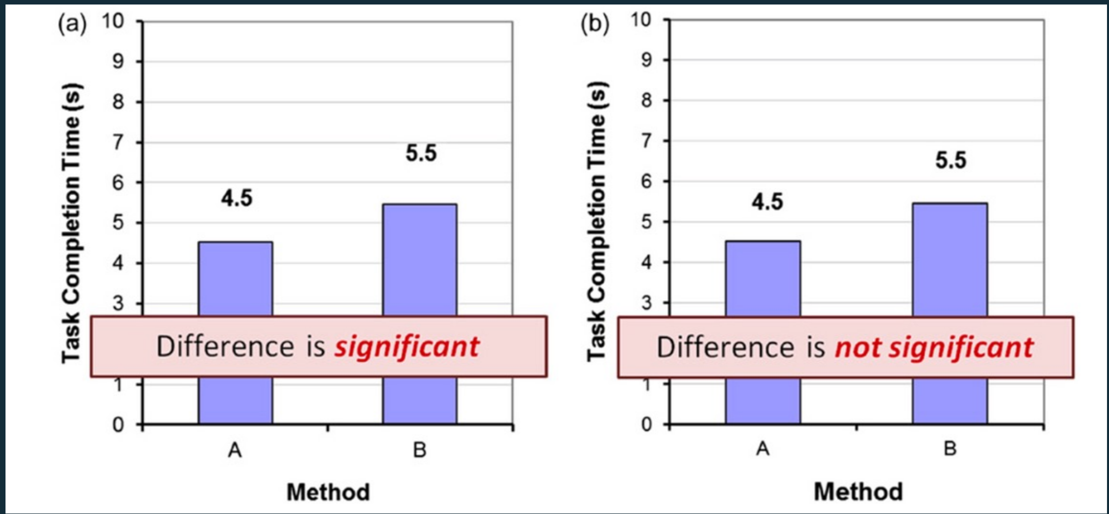
Inferential Statistics

Infer something about the population based on the sample.



Hypothesis Testing

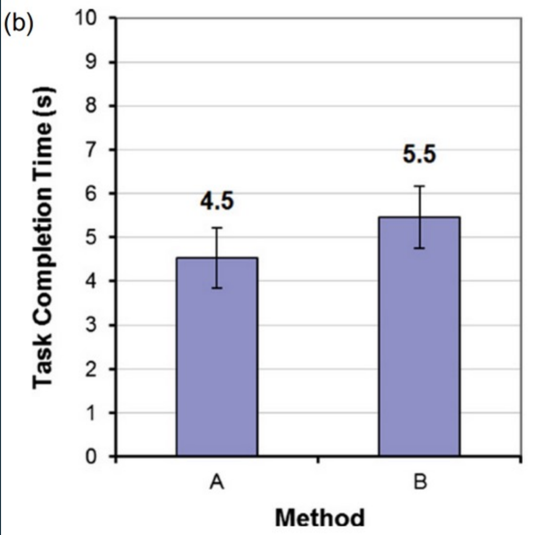
Is method A faster than method B?

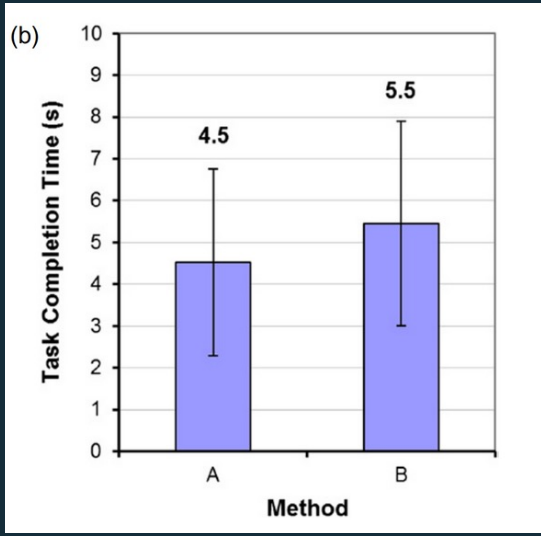


Comparing means/medians is not enough!

Hypothesis Testing

Is there a difference between method A and B?





Comparing means/medians is not enough!

Statistical Significance

How likely is it that the effect happened by chance?

We quantify this likelihood as p & we decide on an acceptable % called α (usually 5% or 1%).

When $p < 0.05$ → the difference is likely not caused by chance
→ method impacts task completion time

Null & Alternative Hypothesis

H1: Method A is faster than method B.

H0: There is no difference in speed between method A & B.

When $p < 0.05$, we reject H0 → evidence for H1

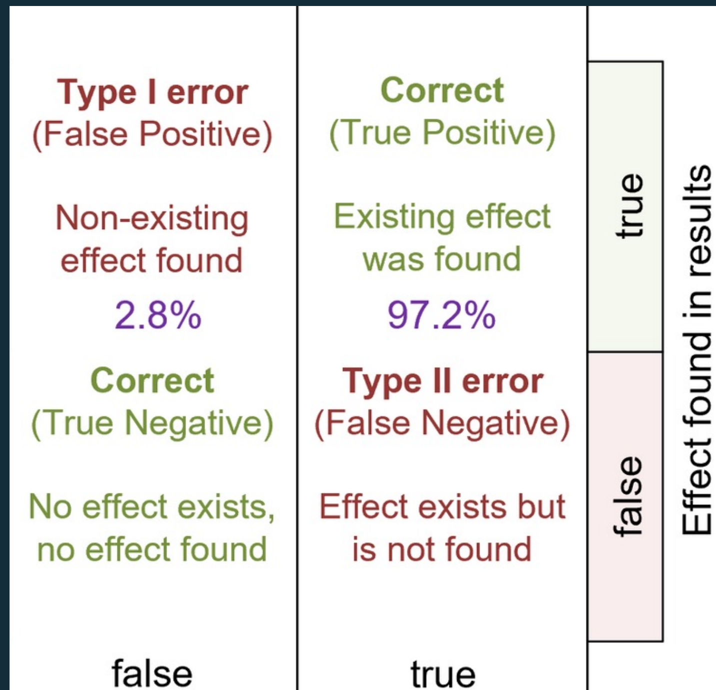
When $p > 0.05$ we fail to reject H0 → insufficient evidence for H1

not we accept H0!

Null & Alternative Hypothesis

H1: There is a difference

H0: There is no difference



Hypothesis Testing

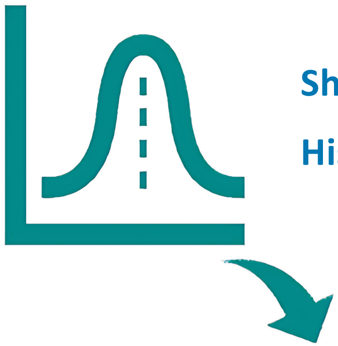
What is your experimental design?

How many IV levels do you have?

What is the data type? Is the data normally distributed?

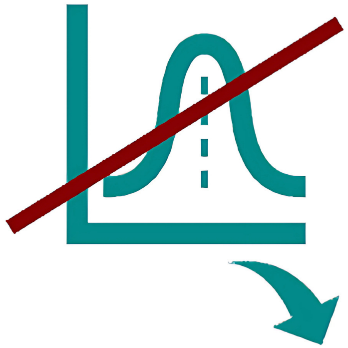
The Right Test

Parametric or not?



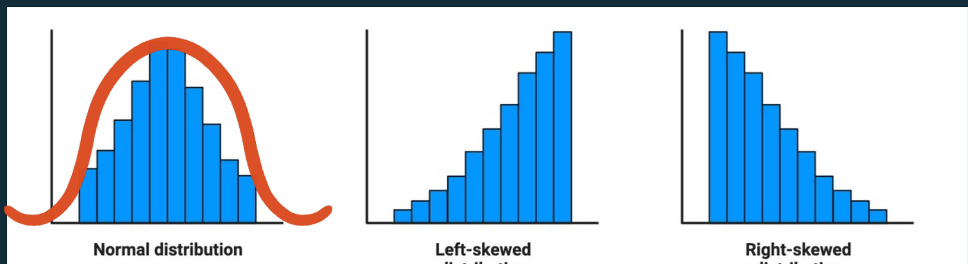
Shapiro-Wilk test

Histograms

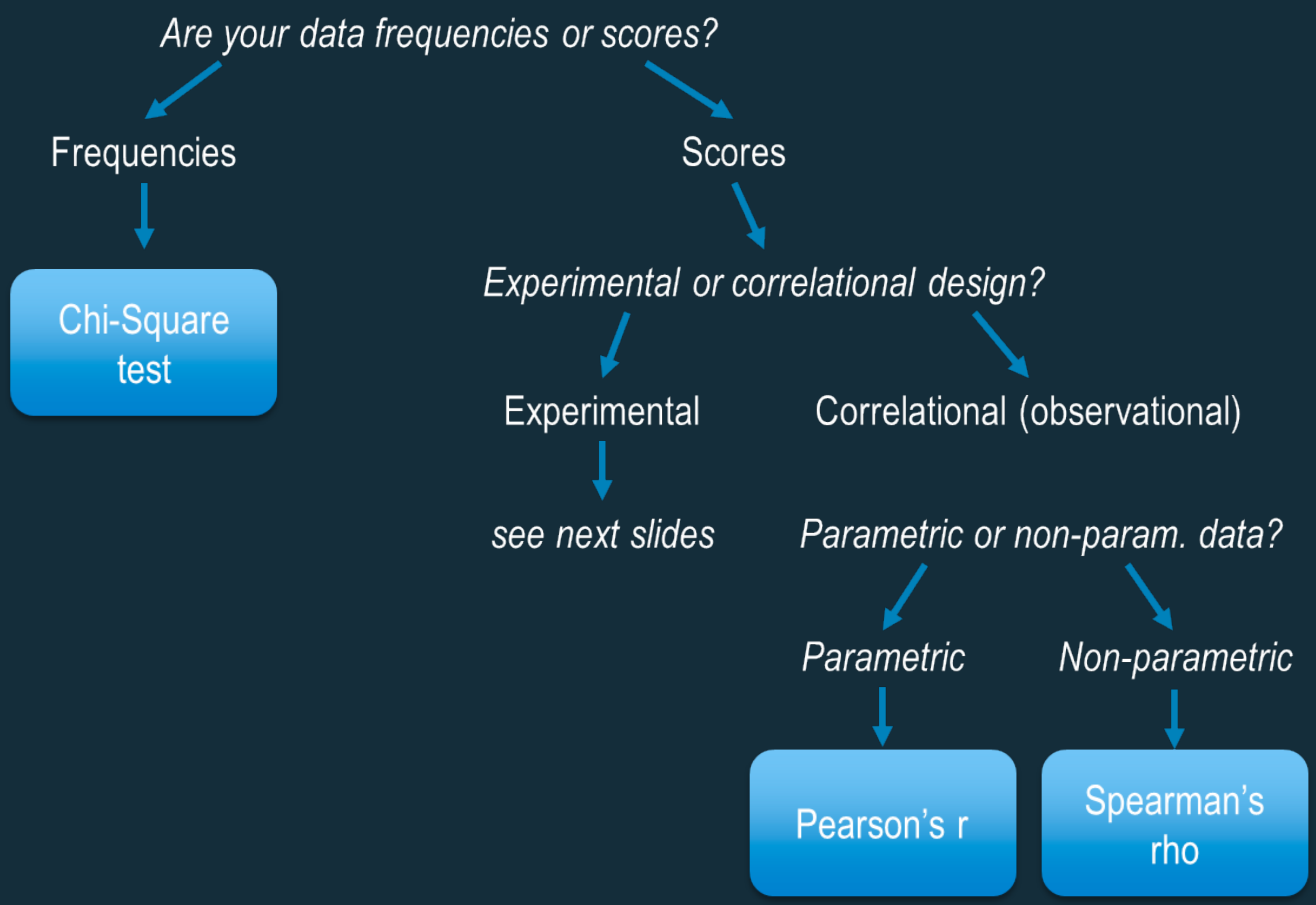


Data normally distributed, then **parametric tests** are used.

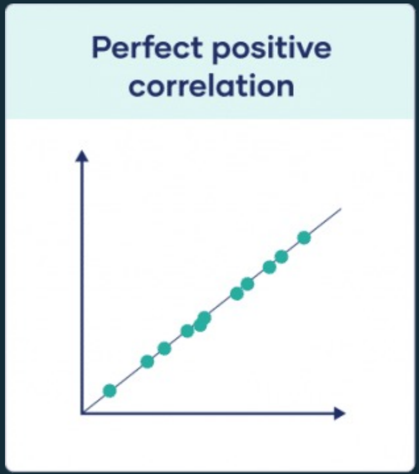
Data not normally distributed, then **non-parametric** tests are used.

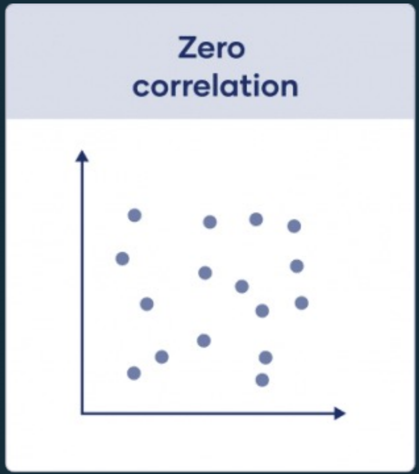


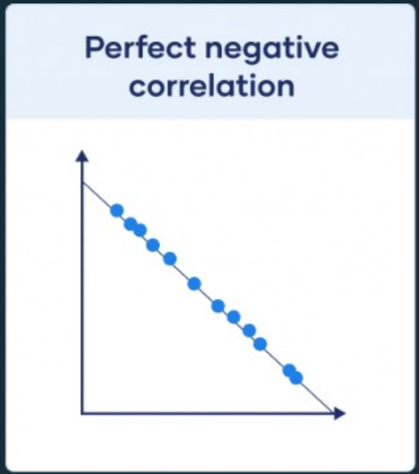
Choosing the right test



Correlations



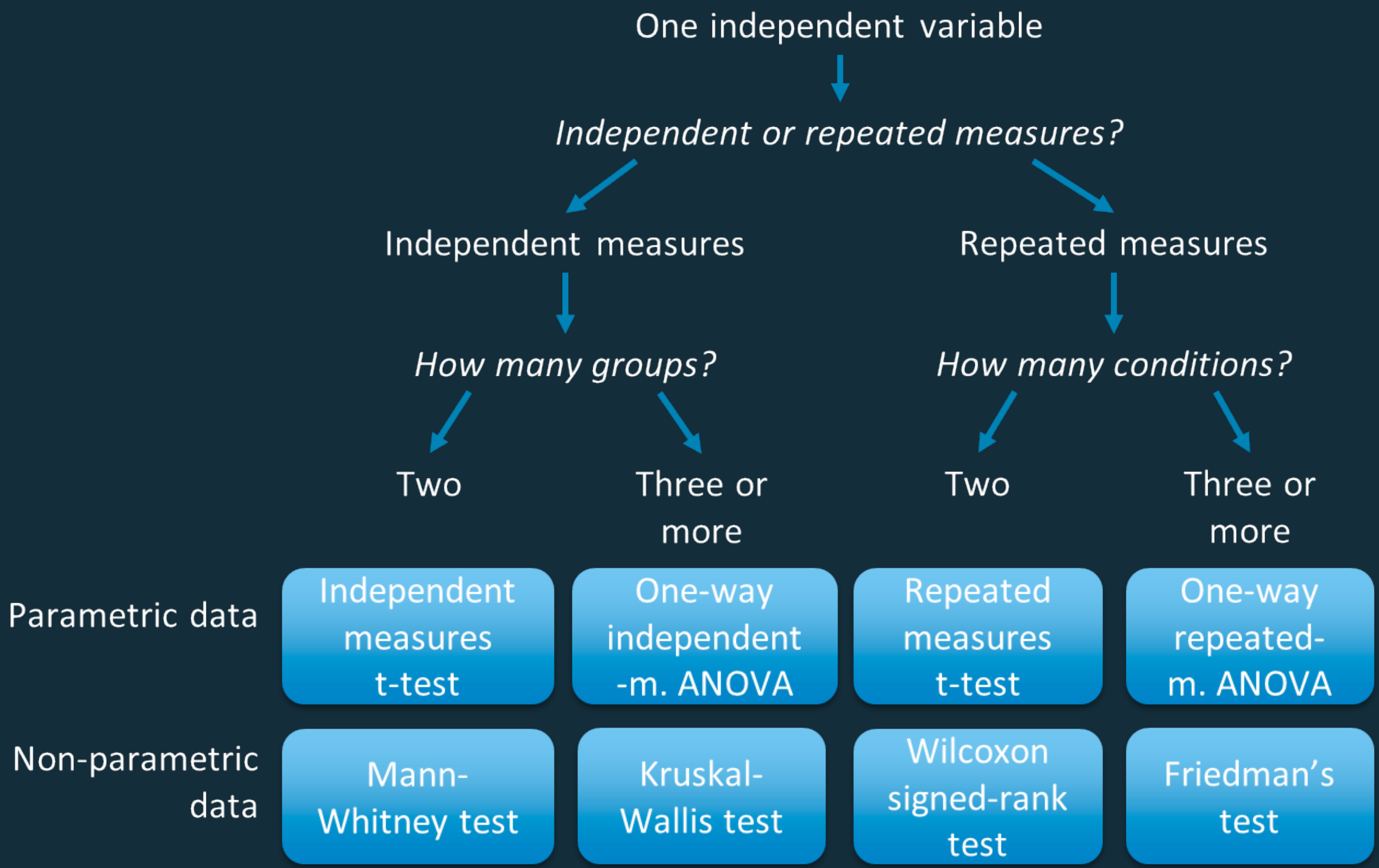




Pearson's correlation coefficient

	Coefficient, *r*	
Strength of Association	Positive	Negative
Small	.1 to .3	-0.1 to -0.3
Medium	.3 to .5	-0.3 to -0.5
Large	.5 to 1.0	-0.5 to -1.0

Experimental designs

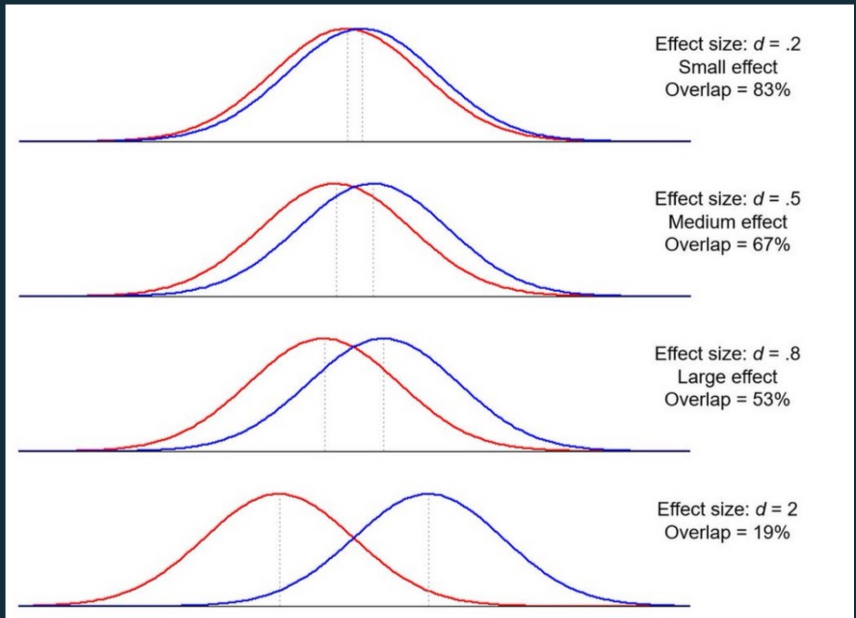


Overview Table

Data Setup	Parametric test	Non-Parametric test
1 Variable 2 Categories Between Subjects	independent t-test	Mann-Whitney U test
1 Variable 2 Categories Within-Subjects	paired t-test	Wilcoxon Signed Rank Test
1 Variable >2 Categories Between Subjects	One-way ANOVA	Kruskal Wallis Test
1 Variable >2 Categories Within Subjects	repeated measures ANOVA	Friedman test Mood's median test
1 Variable (Correlation)	Pearson's r	Spearman's ρ (rho)

Effect Size

We have an effect but how big is it?



Cohen's d:

- d >= 0.20 (small effect)
- d >= 0.50 (medium effect)
- d >= 0.80 (large effect)

Effect Size

Effect Size	Analysis
Cohen's *d*	Independent samples *t* test
Odds ratio (*OR*)	Logistic regression
Pearson correlation (*r*)	Correlation analysis
R squared (R^2)	Multiple regression
Omega squared (ω^2)	ANOVA

Examples

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Exercise

1. Think of a research question (HCI)
2. What data will you collect and what type is it?
3. What is your H0 and H1?
4. What type of test will you use?
    a. In case the data is normally distributed
    b. In case it is not

Tools

SPSS, STATA, R, PYTHON, JASP

*R and Python are recommended as they are free and offer powerful tools for data visualization.

Useful packages/libraries for data visualization and analysis:

R: dplyr, ggplot2, tidyr

Python: Pandas, Plotly, SciPy, Matplotlib

Python

Installing User Guide API reference Building from source

SciPy documentation

Date: April 03, 2024 **Version**: 1.13.0

Download documentation: https://docs.scipy.org/doc/

Useful links: Install | Source Repository | Issues & Ideas | Q&

SciPy (pronounced "Sigh Pie") is an open-source software f

User guide

scipy.stats.ttest_rel

```
scipy.stats.ttest_rel(a, b, axis=0, nan_policy='propagate', alternative='two-sided', *, keepdims=False)
```
[source]

Calculate the t-test on TWO RELATED samples of scores, a and b.

This is a test for the null hypothesis that two related or repeated samples have identical average (expected) values.

Parameters:

a, b : *array_like*

The arrays must have the same shape.

axis : *int or None, default: 0*

If an int, the axis of the input along which to compute the statistic. The statistic of each axis-slice (e.g. row) of the input will appear in a corresponding element of the output. If `None`, the input will be raveled

R

Rdocumentation | powered by datacamp

stats (version 3.6.2)

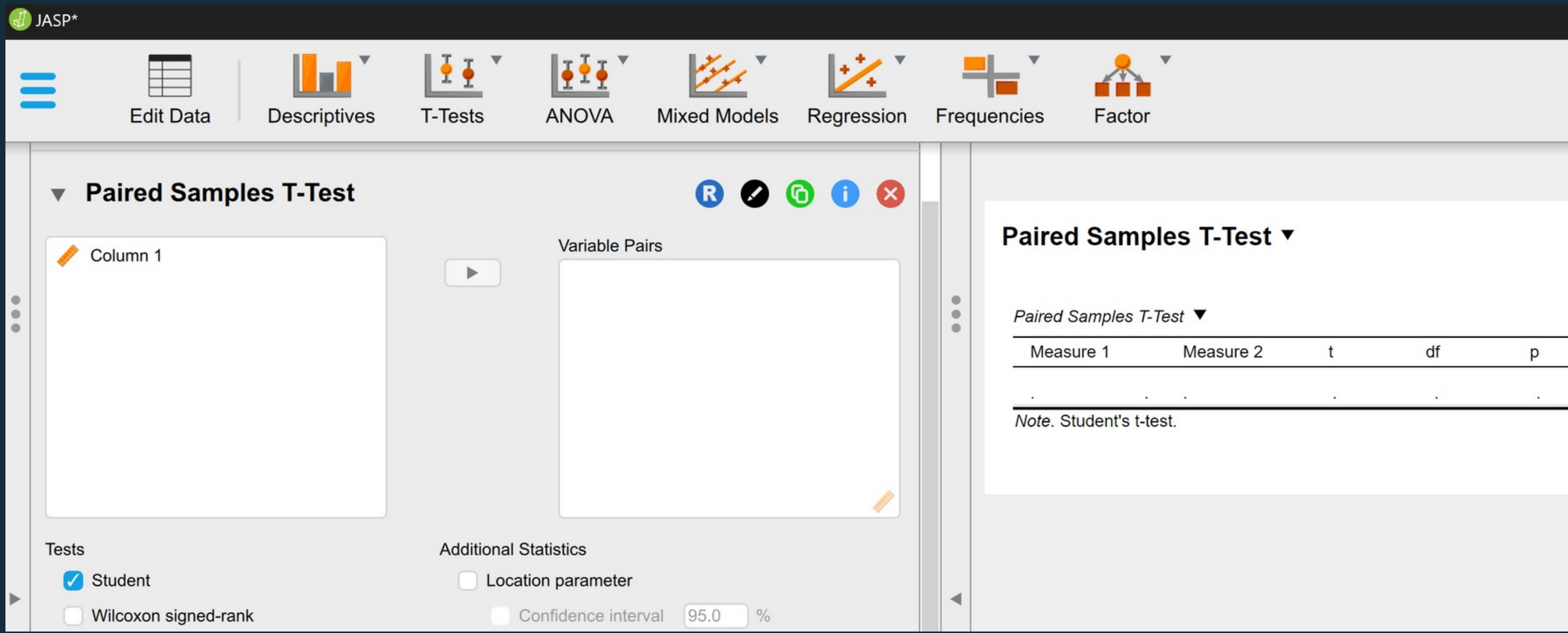
t.test: Student's t-Test

```
t.test(x, y,
      alternative = c("two.sided", "less", "greater"),
      mu = 0, paired = FALSE, var.equal = FALSE,
      conf.level = 0.95, ...)
```

Explain code

POWERED BY datalab

JASP


JASP*
Edit Data
Descriptives
T-Tests
ANOVA
Mixed Models
Regression
Frequencies
Factor
Paired Samples T-Test
Column 1
Variable Pairs
Tests
Student
Wilcoxon signed-rank
Additional Statistics
Location parameter
Confidence interval
95.0
%
Paired Samples T-Test
Paired Samples T-Test
Measure 1
Measure 2
t
df
p
Note. Student's t-test.

Reporting Results

APA style

"A ***paired-samples t-test*** was conducted to compare (DV measure) _________ in (IV level / condition 1) ________and (IV level / condition 2)________ conditions.

There was ***a significant (not a significant)*** difference in the scores for IV level 1 (M=___, SD=___) and IV level 2 (M=___, SD=___) conditions; t(__)=___, p = _____"

"There was a significant difference in the scores for caffeine (M=5.4, SD=1.14) and no caffeine (M=9.4, SD=1.14) conditions; t(4)=-5.66, p = 0.005."

Paired Samples Test

	Paired Differences					t	df	Sig. (2-tailed)
	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference: Lower	95% Confidence Interval of the Difference: Upper			
Pair 1 CAFDTA - NOC	-4.0000	1.5811	.7071	-5.9632	-2.0368	-5.657	4	.005

Visualization

Tables & Figures

Tables: grouping detailed data

Repeated Measures ANOVA ▼

Metric 1 Analysis across conditions between cohorts

Within Subjects Effects

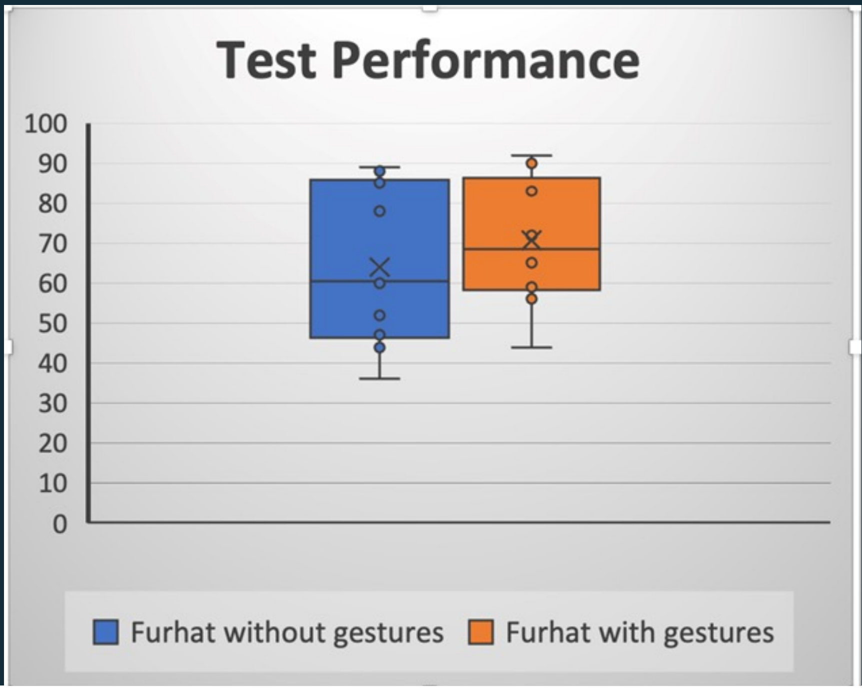
	Sum of Squares	df	Mean Square	F	p	η^2
Conditions	17.213	1	17.213	580.083	< .001	0.706
Conditions ✻ Cohort	0.021	1	0.021	0.720	0.397	0.001
Residual	7.151	241	0.030			

Note. Type III Sum of Squares

Visualization

Tables & Figures

Figures: showing patterns & relationships in data

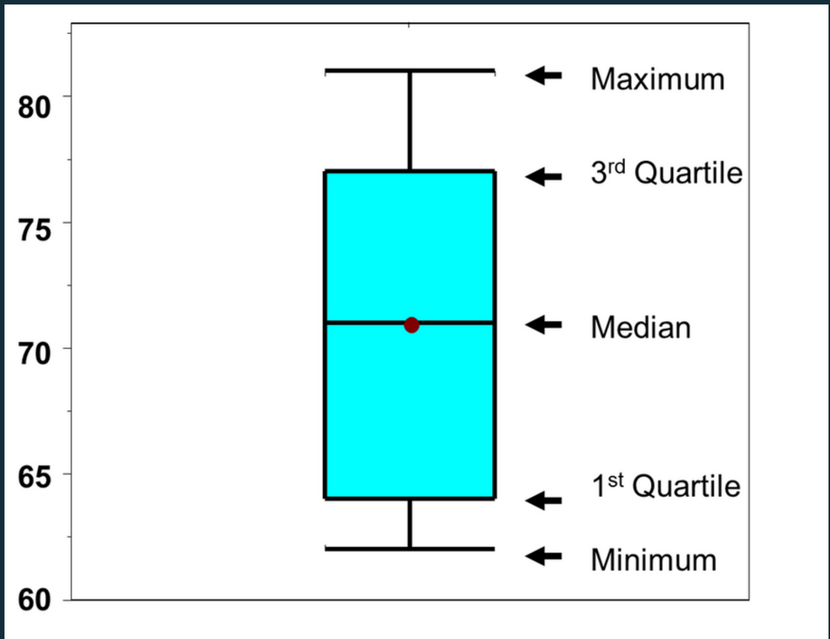


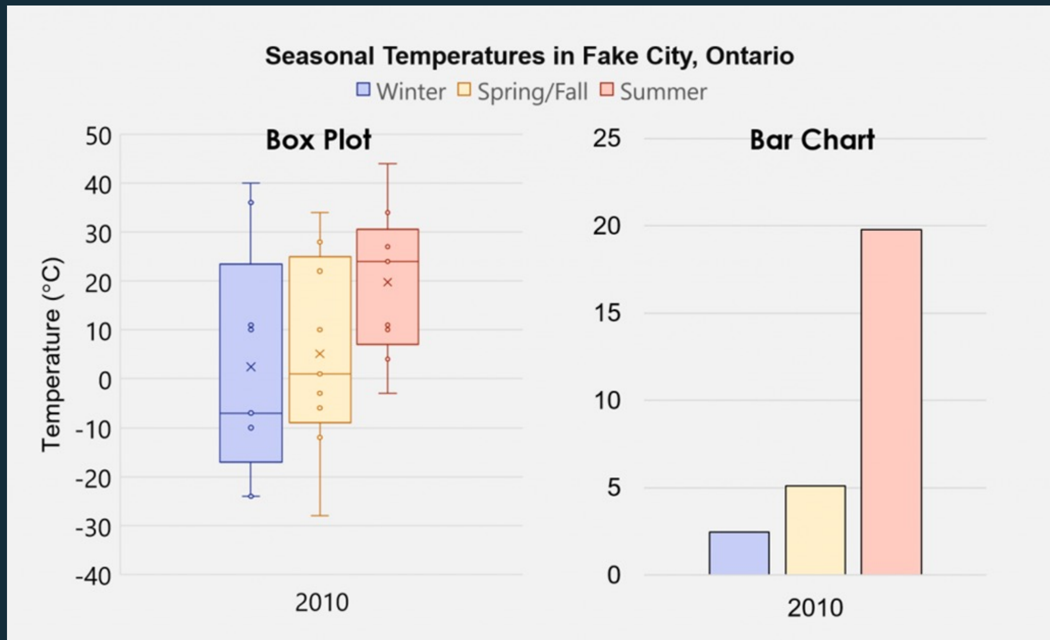
	Test performance	
Subject ID	Furhat without gestures	Furhat with gestures
1	52	59
2	47	61
3	89	83
4	78	90
5	85	92
6	44	56
7	61	72
8	60	85
9	88	65
10	36	44

Figures

Box plots vs bar plots

More informative than bar plots → recommended





Figures

Essential parts

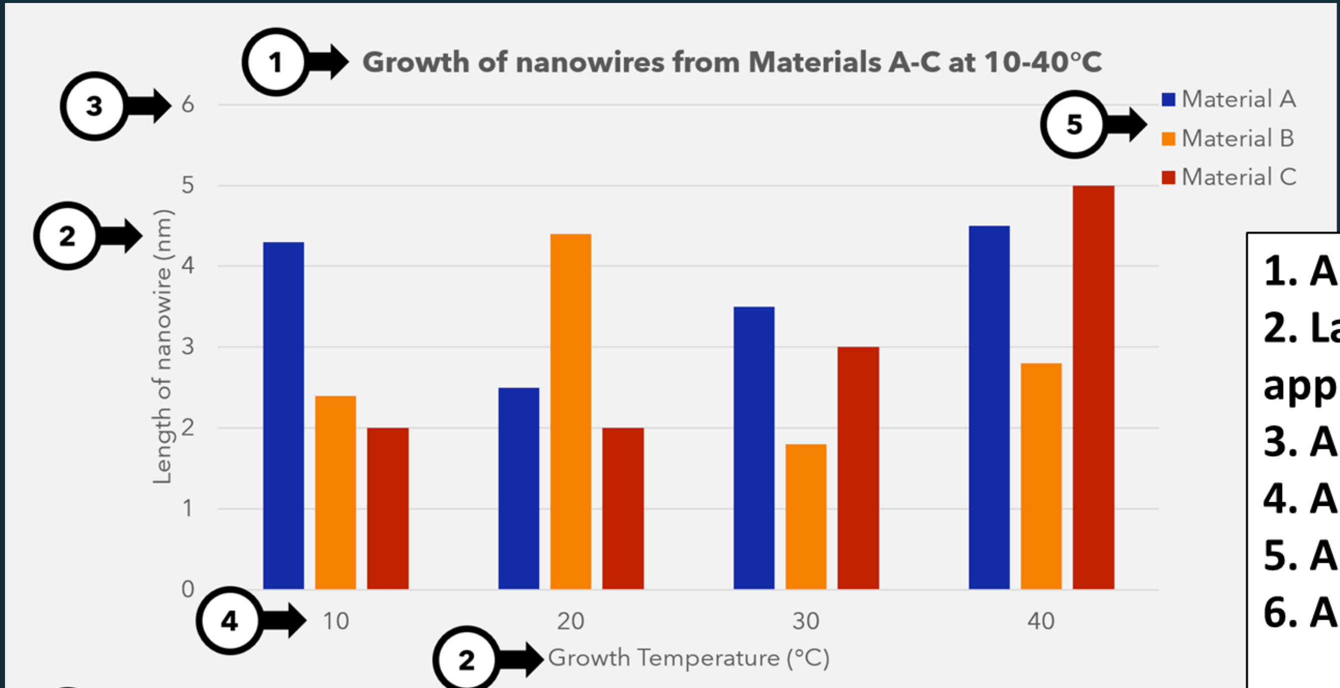
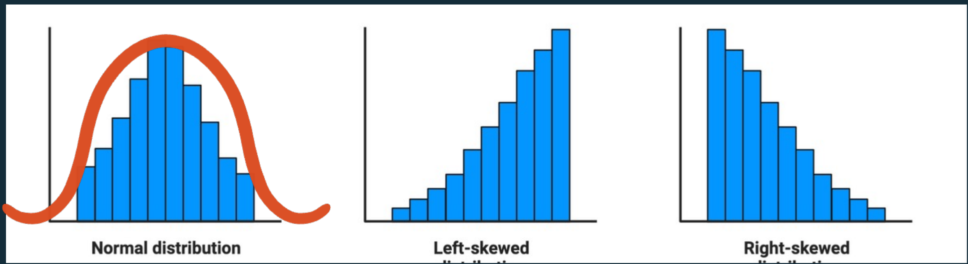


Fig. 1. Measurements in nanometers (nm) of nanowire grown from Materials A, B, and C at 10 to 40 C. All samples prepared following the standard procedure and measured using TEM.

1. A title
2. Labelled axes with units if applicable
3. Appropriate axes scales
4. Appropriate precision
5. A legend
6. A caption

Main Takeaways

- What you need to know before choosing a test:
  - What kind of data do you have?
  - Is it normally distributed?



Shapiro-Wilk test

Histograms

Main Takeaways

- You always have a H0 and H1
  - $p < 0.05$ → reject H0
  - $p > 0.05$ → fail to reject H0

< 5% chance of observing your results (or more extreme) if H_0 were true → evidence for H1